

Overlapping Generations in Lean: The Life-Cycle Household

LeanEconomics Org [mostly Claude]*

September 2026

Draft, in progress

Abstract

This is the record of a programme to formalise overlapping-generations economies in Lean 4, and of its first stage: the life-cycle household. A household that lives J periods against uninsurable earnings risk and a borrowing constraint solves a finite backward induction, and everything proved for the infinite-horizon Bellman operator with an arbitrary continuation applies stage by stage with no fixed point. Beyond that we prove the finite-horizon sandwich: with k periods still to come, consumption lies between $\kappa_k m$ and $\kappa_k(m + H_k(z))$, where $1/\kappa_k = \sum_{i \leq k} (\mathfrak{p}/R)^i$ is the consumption share of a household with no future income and H_k is the expected present value of the next k incomes; the propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \mathfrak{p}/R$. The programme's target is uniqueness of the stationary equilibrium at relative risk aversion above one, which the infinite horizon could not close: in a life cycle every comparative static is a finite induction, wealth is bounded, and the age-one distribution is common to every rate.

JEL: C63, D52, E21. *Keywords*: overlapping generations, life cycle, Bewley models, formal verification, Lean.

1 Introduction

The companion write-up in this repository proved that the stationary equilibrium of Aiyagari (1994) is unique at his logarithmic calibration, and located exactly what stands in the way for relative risk aversion $\mu \in \{3, 5\}$. Every obstacle it met traces to the fixed point of the infinite-horizon problem: a condition on the response of consumption to the interest rate fails in a tail of wealth that the stationary distribution reaches; an induction that would prove the response bound has a step multiplier $1/\mathfrak{p} > 1$, so it cannot close as a supremum argument; and the only floor on capital supply comes from an ergodic identity that loses

*The Lean development described here is at <https://github.com/LeanEconomics/LeanEconomics>. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. This is the companion to the write-up on Aiyagari (1994) in the same repository. Comments welcome.

a factor $\|u'\|^2 / \text{Var}$. Overlapping generations remove all three at once. A household that lives J periods solves a finite backward induction from an explicit terminal condition, so a multiplier $1/p$ per step compounds to p^{-J} , which at Aiyagari's numbers is about 1.03. Wealth is bounded by J periods of accumulation, so any condition needs checking on a known bounded region. And the distribution of the youngest cohort is common to every interest rate, so an ordering of policies pushes forward to an ordering of the whole age profile of distributions by induction on age, with no Doeblin atom and no uniqueness of a stationary distribution.

The target of the programme is uniqueness of the stationary equilibrium of a stochastic life-cycle economy in the manner of Huggett (1996) at $\mu \in \{3, 5\}$. Section 2 records the first stage, the life-cycle household, which is formalised. Section 3 records the rest of the programme, the classical results of Diamond (1965) included, and what each stage needs. As in the companion, every result stated with a formal name has been checked by the Lean kernel against Mathlib; the development is at <https://github.com/LeanEconomics/LeanEconomics>, directory `LeanEconomics/OLG`.

2 The life-cycle household

2.1 Stages

The household of the companion write-up is kept: earnings $y(z)$ follow a finite Markov chain π , the gross return is $R = 1 + r$, cash on hand is $m = Ra + y(z)$, saving $a' \in [0, \bar{a}]$ is bounded below by the borrowing limit and above by an artefactual cap, utility is CRRA with $u'(c) = c^{-\mu}$ and $\beta \in (0, 1)$. What changes is the horizon: the household lives J periods and leaves nothing behind. Count periods from the end. With k periods still to come after today, the continuation value is the k -th Bellman iterate from zero,

$$V_k = T^k 0, \quad (Tv)(a, z) = \max_{a'} \{u(m - a') + \beta \sum_{z'} \pi(z, z') v(a', z')\},$$

and the household with k periods to come saves $g_k = \text{policyOf } V_k$ and consumes $c_k = m - g_k$ (`stageValue`, `stagePolicy`, `stageConsumption`). Age j of a J -period life is stage $k = J - 1 - j$. In the last period the household saves nothing and consumes its cash on hand (`stagePolicy_zero`, `stageConsumption_zero`).

This is the finite-horizon problem with a flat age profile of earnings: the same income process at every age. An age profile, retirement and survival risk need income to be a parameter of the Bellman step rather than a field of the structure, and are deferred to the next stage of the programme.

2.2 What transfers without a fixed point

The companion development proved its household results for the Bellman operator applied to an arbitrary continuation v , and only then passed to the fixed point. Here there is no fixed point to pass to, and each result is read at $v = V_k$:

- consumption is positive at every stage when marginal utility is unbounded at zero (`stageConsumption_pos`);
- the stage value has concave slices, so the optimal action is unique and both saving and consumption rise with wealth (`stagePolicy_mono`, `stageConsumption_mono`);
- Carroll–Kimball: consumption is concave in wealth at every stage (`concaveOn_stageConsumption`);
- the Euler inequality under the cap, $\beta R \mathbb{E}[u'(c_k(a', z')) \mid z] \leq u'(c_{k+1}(a, z))$, at every state (`stage_euler_le`), and the Euler inequality above the floor, the reverse, at every state where the household saves (`stage_euler_ge`).

2.3 The finite-horizon sandwich

Let $\mathfrak{p} = (\beta R)^{1/\mu}$ and define the finite-horizon minimal propensities to consume by

$$\kappa_0 = 1, \quad \kappa_{k+1} = \frac{\kappa_k R}{\mathfrak{p} + \kappa_k R}, \quad \text{so that} \quad \frac{1}{\kappa_k} = \sum_{i=0}^k \left(\frac{\mathfrak{p}}{R}\right)^i$$

(`stageMPC`): the share of cash on hand consumed by a household with k periods to come and no future income. Let $H_0 = 0$ and $RH_{k+1}(z) = \sum_{z'} \pi(z, z')(y(z') + H_k(z'))$, the expected present value of the next k incomes (`stageHumanWealth`).

Theorem 2.1 (The finite-horizon sandwich). *Assume CRRA utility with marginal utility unbounded at zero, and that the cap is slack at every stage. Then for every k , every z and every $a \in [0, \bar{a}]$,*

$$\kappa_k m \leq c_k(a, z) \leq \kappa_k (m + H_k(z)).$$

Formal statement: `IncomeFluctuation.stageMPC_mul_le_stageConsumption`,
`IncomeFluctuation.stageConsumption_le_stageHumanWealth`.

Each bound is one Euler step, applied k times. The lower bound: if consumption against v is at least κm , the Euler inequality above the floor at a state where the household saves gives $c^{-\mu} \leq \beta R \mathbb{E}[(\kappa R a')^{-\mu}]$ with $a' = m - c$, hence $\kappa R(m - c) \leq \mathfrak{p}c$ and $c \geq \kappa R(\mathfrak{p} + \kappa R)^{-1}m$; at a corner $c = m$ (`ccra_stageMPC_step`). The upper bound: if consumption against v is at most $\kappa(m + H(z))$, the Euler inequality under the cap and Jensen for the negative power give $c^{-\mu} \geq \beta R (\mathbb{E}[c'])^{-\mu}$ with $\mathbb{E}[c'] \leq \kappa R(a' + H'(z))$, hence $c \leq \kappa R(\mathfrak{p} + \kappa R)^{-1}(m + H'(z))$ (`ccra_stageUpper_step`). The recursion for κ is the same on both sides, which is why the sandwich is exact in the homogeneous case.

The propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \mathfrak{p}/R$ of the companion write-up: $\kappa \leq \kappa_k$ for every k (`minMPC_le_stageMPC`), so a household with less life left consumes a larger share of its cash, and the infinite-horizon bounds are the limiting case. A first consequence for the programme is an explicit bound on saving, $g_k(a, z) \leq (1 - \kappa_k)(Ra + y_{\max})$ (`stagePolicy_le`), which iterated from zero wealth at the start of life bounds the wealth of every cohort.

3 The programme

The stages, in the order we intend to take them. The first is done (Section 2).

Equilibrium. A stationary overlapping-generations equilibrium: the distribution μ_1 of the youngest cohort is given (zero wealth, earnings drawn from the invariant law), μ_{j+1} is the pushforward of μ_j under the age- j policy and the earnings chain, capital supply is $K(r) = \sum_j \omega_j \int a d\mu_j$ with cohort weights ω_j , labour is exogenous, and the firm is Aiyagari's. Targets: continuity of K in the rate, a finite composition of continuous pushforwards; existence by the intermediate value theorem against the firm's demand, with the floor at the top of the interval from explicit accumulation over the life rather than from an ergodic identity; Light's Theorem 2 by induction on age, an ordering of policies at every age giving first-order stochastic dominance of every μ_j and so of K ; and single crossing against a decreasing demand.

Theorem 1 by backward induction. The research core: saving rises with the rate at every age, for $\mu \in \{3, 5\}$. The reduction of the companion write-up applies stage by stage, Theorem 1 at stage $k + 1$ following from the slack elasticity condition on the stage- k consumption functions at the pivot; the transfer lemma propagates an affine bound on the response of consumption to the rate backward from the terminal condition $c_0 = m$, where the response is exactly $\Delta r \cdot a$; and the slack condition at each stage is a finite check on the constants so obtained. The obstruction of the infinite horizon, a step multiplier of at least $1/p$, does not bite in J steps. The first session is numerical: a life-cycle economy with $J = 60$, Aiyagari's earnings chain and $\mu \in \{3, 5\}$, checking the slack condition at every age and wealth and running the recursion for the constants. If it holds the argument is formalised; if it fails only at moderate ages and wealth we look for the age-dependent refinement; if it fails badly, log utility is the fallback, where Theorem 1 at each stage follows from the rescaling proof of the companion, which is already an induction on Bellman iterates.

Diamond (1965) and Samuelson. The two-period economy without risk: a unique steady state and monotone global convergence for logarithmic utility and Cobb–Douglas production, existence under Inada conditions and a multiplicity witness in general, dynamic inefficiency of a steady state with $r < n$ by an explicit Pareto-improving transfer, the golden rule, and the autarkic and monetary steady states of the pure-exchange economy. These are standalone results and will be the classical sections of this write-up.

Extensions. An age profile of earnings, retirement with a pension and a government budget, survival risk with accidental bequests as a second fixed point, population growth, and Lasry–Lions with age as time, whose finite-horizon core is already formalised and whose monotonicity is again the elasticity condition.

References

- AIYAGARI, S. R. (1994): “Uninsured Idiosyncratic Risk and Aggregate Saving,” *Quarterly Journal of Economics*, 109, 659–684.
- DIAMOND, P. A. (1965): “National Debt in a Neoclassical Growth Model,” *American Economic Review*, 55, 1126–1150.
- HUGGETT, M. (1996): “Wealth Distribution in Life-Cycle Economies,” *Journal of Monetary Economics*, 38, 469–494.

A Index of formal results

All in the namespace `LeanEconomics.IncomeFluctuation`, directory `LeanEconomics/OLG`. Every result depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and none uses `sorry`.

Result	Lean name	Module
Stage value, policy, consumption	<code>stageValue,</code> <code>stageConsumption</code>	<code>stagePolicy,</code> OLG/LifeCycle
Last period: no saving	<code>stagePolicy_zero</code>	OLG/LifeCycle
Last period: consume cash on hand	<code>stageConsumption_zero</code>	OLG/LifeCycle
Positive consumption at every stage	<code>stageConsumption_pos</code>	OLG/LifeCycle
Saving and consumption rise with wealth	<code>stagePolicy_</code> <code>stageConsumption_mono</code>	<code>mono,</code> OLG/LifeCycle
Carroll–Kimball at every stage	<code>concaveOn_stageConsumption</code>	OLG/LifeCycle
Euler under the cap, by stage	<code>stage_euler_le</code>	OLG/LifeCycle
Euler above the floor, by stage	<code>stage_euler_ge</code>	OLG/LifeCycle
Finite-horizon propensities, human wealth	<code>stageMPC, stageHumanWealth</code>	OLG/LifeCycle
Propensities in $(0, 1]$, above κ	<code>stageMPC_pos,</code> <code>stageMPC_le_one,</code> <code>minMPC_le_stageMPC</code>	OLG/LifeCycle
Lower Euler step	<code>crra_stageMPC_step</code>	OLG/LifeCycle
Upper Euler step	<code>crra_stageUpper_step</code>	OLG/LifeCycle
Theorem 2.1, lower	<code>stageMPC_mul_le_stageConsumption</code>	OLG/LifeCycle
Theorem 2.1, upper	<code>stageConsumption_</code> <code>stageHumanWealth</code>	<code>le_</code> OLG/LifeCycle
One-step saving bound	<code>stagePolicy_le</code>	OLG/LifeCycle